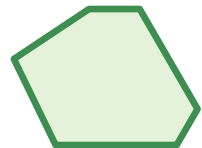


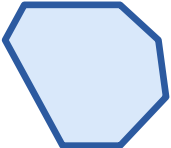
Constraints

Wrench Ineq.



Eq. (26)

Kinematic Ineq.

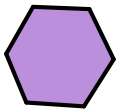
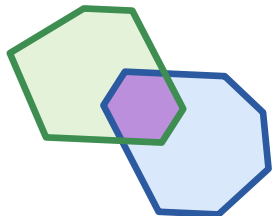


Eq. (32)



Admissible Set

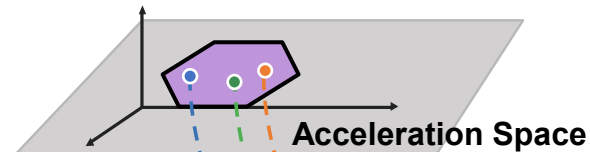
Unified as a Single Admissible Set in Acceleration Space



Eq. (34)

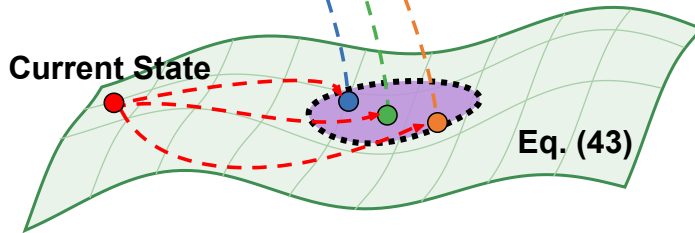
State Mapping on Manifold

Projected Accelerations Mapped onto the Configuration Manifold



Acceleration Space

Current State

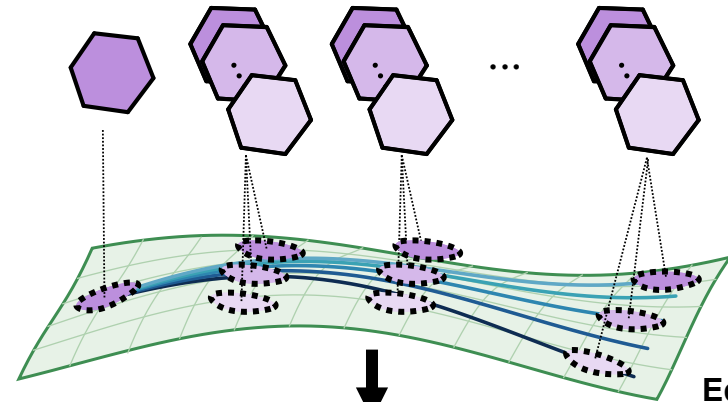


Eq. (43)

Manifold in Configuration Space

Constrained Rollout

Constrained Propagation and Cost Evaluation over the Prediction Horizon



Eq. (46)

Execution

Feasibility-Preserving Weighted Control Update Eq. (51)